

abhishekh reddy munnangi

graduate student in robotics

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education

M.Eng in Robotics at University of Maryland, College Park

Aug 2022 - May 2024 (Expected)

Finished Third Semester with 3.33 GPA

B.Tech in Mechatronics Engineering at Ajeenkya DY Patil University, Pune

Aug 2018 - May 2022

Graduated with 7.87/10 CGPA

skills

Programming Languages	C++ and Python
Software Development Tools	Git, Docker, Shell Scripts, CMake and GoogleTest
Frameworks and Libraries	ROS2 (rclcpp), MoveIt!, OpenCV, Orocos-KDL, OpenGL, Eigen and NumPy
Documentation	Doxygen, Sphinx, Markdown and LaTeX

experience

Graduate Assistant at Institute for Systems Research, UMD

Aug 2023 - Present

- Served as a Teaching Assistant for a Robotics Laboratory Course: ENEE467 in Fall '23 semester. Worked with students from diverse engineering backgrounds.
- Guided students in interdisciplinary lab exercises which included writing a roscpp node to draw shapes with a real UR3e robotic arm and developing a line follower program for the TI Robotic Systems Learning Kit.
- Delivered a hotfix for the C++ wrapper library based on MoveIt! API used during the course. This significantly improved execution success rate of student-written code when using the Cartesian motion planner to move the robotic arm.
- Migrated the documentation for lab exercises from GitHub Wiki to Sphinx and hosted it on Read The Docs, resulting in centralized documentation for improved organization and maintenance.
- Assisted students with setting up GPIOs, SysTick timer, General purpose timer and PWM signal generation on TI-MSP432 Launchpad board.

Robotics Engineer Intern at iTrontik Smart Systems, Pune

May 2021 - Apr 2022

- Sized the components of the quadcopter drive system for hover flight time and selected suitable off-the-shelf parts.
- Designed a sturdy 3D-printed quadcopter frame for prototyping using Fusion 360 in three iterations. The frame was able to sustain multiple hard and crash landings during testing.
- Analyzed flight logs to tune the attitude PID controller and mitigated vibrations in the prototype frame.
- Ideated and developed a USB-C charger board which integrates into the battery pack to reduce the need to ship an external charger with the drone. Designed a 2-layer PCB in KiCAD and worked with a PCB fab house to produce the first prototype.

projects

Justbot Manipulator as coursework project

Nov 2022 - Dec 2022

- Simulated kinematics of a 7 DOF articulated serial manipulator designed in Solidworks using Gazebo and ROS.
- Written an inverse kinematics solver in Python that publishes joint angles for a given end-effector displacement.

Mint Flight Controller Board as personal project

Oct 2020 - May 2021

- Designed a miniature flight controller board for quadcopter drones in KiCAD based on STM32H7 microcontroller.
- The resulting 4-layer PCB sizes 80 percent smaller than Pixhawk 4 while retaining most functionality.

Brushless Motor Speed Controller as coursework project

Nov 2019 - Dec 2019

- Implemented brushless motor control using a 3-phase inverter circuit with MOSFETs and gate drivers.
- Created the circuit schematic in KiCAD and hand-soldered the components on a perfboard.